

A Comparative Study of Positional Encoding Methods for Length Extrapolation

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Abstract—Transformer models rely on self-attention to process input sequences. Self-attention does not preserve token order, so positional encoding supplies sequence information during training and inference. The choice of positional encoding influences representation quality, optimization, and generalization to unseen sequence lengths. This property has become increasingly important because modern language models often process contexts much longer than their training window. Several positional encoding methods have been proposed to address this limitation, including absolute, relative, rotary, and bias-based approaches. This paper surveys the broader landscape of positional encoding research and then narrows its comparative focus to three representative methods spanning the principal design paradigms: Learned Absolute Positional Embedding, Rotary Positional Encoding (RoPE), and Attention with Linear Biases (ALiBi). The comparison focuses on theoretical motivation, computational cost, implementation complexity, and length extrapolation performance reported in recent literature.

Index Terms—Transformer, Positional Encoding, Length Extrapolation, Self-Attention, RoPE, ALiBi, Large Language Models

I. INTRODUCTION

Transformer architectures have become the standard architecture for natural language processing and increasingly appear in speech, vision, and multimodal learning tasks [1], [2]. Their success comes from the self-attention mechanism, which models long-range dependencies while supporting efficient parallel computation. Self-attention alone does not preserve token order. A transformer therefore treats an input sequence as an unordered collection unless positional information is introduced [1].

Positional encoding addresses this limitation by assigning each token a representation of its position within the sequence. This information allows the attention mechanism to distinguish identical tokens appearing at different locations. The positional representation influences attention scores, contextual representations, and downstream model performance. As a result, positional encoding has become a fundamental component of transformer architectures [1], [7].

Early transformer models relied on absolute positional encodings. Two common variants emerged. The first used fixed sinusoidal functions, while the second learned position embeddings during training [1], [3], [4]. Both approaches performed well within the training context window. Their performance declined once the input sequence exceeded the

maximum length observed during training. Learned embeddings fail because unseen positions have no corresponding representation. Sinusoidal encodings remain mathematically defined, yet models rarely learn a stable positional structure outside the training distribution [5].

This limitation has become increasingly important with the growth of large language models. Modern applications routinely process long documents, source code, retrieval-augmented prompts, and multi-turn conversations. These tasks often require context windows far beyond the length used during pretraining. Extending the training context directly is computationally expensive because self-attention scales quadratically with sequence length. Memory consumption also increases rapidly as the context grows [5]. Early transformer models typically processed between hundreds and a few thousand tokens, while recent large language models support context windows extending to well over 100K tokens, increasing the importance of reliable length extrapolation [2].

The remainder of this paper is structured to evaluate these developments. Section II reviews the foundational positional encoding paradigms — absolute, rotary, and bias-based — that motivate this study. Section III then narrows the focus to a direct comparison of three representative methods, one from each paradigm: Learned Absolute Positional Embedding, Rotary Positional Encoding (RoPE), and Attention with Linear Biases (ALiBi). These three were selected because each anchors a distinct strategy for injecting positional information into self-attention, making them a natural basis for a controlled comparison.

The main contributions of this paper are summarized as follows.

- Review of the foundational absolute, rotary, and bias-based positional encoding paradigms, establishing the theoretical context for length generalization.
- Direct comparative analysis of Learned Absolute Positional Embedding, RoPE, and ALiBi across length extrapolation capability, computational overhead, and implementation complexity.
- Discussion of the practical trade-offs that determine which of these three methods is appropriate for a given transformer deployment scenario.

II. LITERATURE REVIEW

Research on positional encoding has progressed from absolute representations to methods designed specifically for long-context inference. Early approaches focused on learning positional information within a fixed context window. Later work shifted toward preserving attention stability beyond the training length. Figure 1 introduces the hierarchical structure of these paradigms.

A. Absolute Positional Encodings

The original Transformer introduced sinusoidal positional encoding as a deterministic representation of token position [1]. Each position is encoded using sine and cosine functions with different frequencies. Every dimension corresponds to a unique frequency, producing a continuous representation across the sequence. Since the encoding is generated analytically, positional vectors exist for any sequence length without requiring additional parameters.

Sinusoidal encoding also preserves relative position information through linear transformations. A positional shift produces a predictable rotation within each frequency component. This property initially suggested better generalization beyond the training context [1]. Later studies reported a different behavior. Models trained on fixed sequence lengths rarely learn the underlying rotational structure well enough to support reliable extrapolation. Performance drops once evaluation sequences extend beyond the training window [5].

Several transformer models replaced sinusoidal encoding with learned absolute embeddings. BERT and GPT-2 both adopt this approach [3], [4]. Each position corresponds to a trainable embedding stored in a lookup table. The model learns positional representations directly from data instead of relying on predefined mathematical functions.

Learned embeddings often improve performance within the training distribution because positional representations adapt to the dataset. Their flexibility also creates a limitation. Positions outside the training range have no learned embedding. Models therefore fail when evaluated on longer sequences unless additional embeddings are introduced through retraining or architectural modification [5]. This weakness motivated the search for positional representations that generalize beyond fixed context lengths.

B. Relative Positional Encodings

Relative positional encoding models the distance between tokens instead of their absolute locations. The attention mechanism therefore depends on pairwise offsets rather than fixed sequence indices. This formulation improves translation invariance and reduces dependence on absolute position values.

Shaw et al. incorporated learnable relative position representations directly into the self-attention computation [7]. Each attention score receives an additional representation based on the distance between the query and key tokens. The model therefore learns how attention changes with relative separation instead of memorizing individual positions.

Transformer-XL extended this idea by combining relative positional encoding with segment-level recurrence [8]. Hidden states from previous segments remain available during subsequent computations, allowing information to flow across segment boundaries. This design substantially increases the effective context length while avoiding repeated computation.

The T5 architecture simplified relative positional encoding by introducing learned scalar biases instead of vector representations [6]. Relative distances are grouped into logarithmic buckets before bias values are applied to the attention logits. Nearby positions receive detailed representations, while distant positions share larger buckets. This design reduces memory requirements and improves robustness for unseen sequence lengths. Performance still declines under extreme extrapolation because distant tokens eventually collapse into the same bucket [5].

C. Rotary Positional Encoding

Rotary Positional Encoding, commonly known as RoPE, has become one of the most widely adopted positional encoding methods in modern large language models [2], [9]. Instead of adding positional vectors to token embeddings, RoPE applies a position-dependent rotation directly to the query and key vectors within each attention head.

The rotation introduces positional information while preserving relative distance during the dot-product operation. Attention scores therefore depend on the positional difference between tokens instead of their absolute locations. This formulation combines the simplicity of absolute encoding with the advantages of relative position awareness [9].

RoPE performs well within the training context and supports efficient implementation in transformer architectures. Despite these advantages, performance still degrades once evaluation sequences extend beyond the training distribution [5], [10]. Later analyses attribute this behavior to instability in high-frequency dimensions, where rapid oscillation distorts attention scores over long distances [13]. These observations motivated several methods that modify RoPE without changing the transformer architecture.

D. Attention with Linear Biases

Attention with Linear Biases, abbreviated as ALiBi, introduces positional information through attention scores instead of embedding vectors [5]. Each attention head applies a linear penalty proportional to the distance between tokens. Different heads use different slopes, allowing attention patterns to vary across multiple distance scales.

Unlike absolute embeddings, ALiBi contains no position-dependent parameters tied to a maximum sequence length. The same bias function applies regardless of context size. This property allows models trained on short sequences to process substantially longer inputs without modifying the positional representation.

Press et al. demonstrated that ALiBi achieves strong extrapolation performance while reducing training memory and computation compared with sinusoidal encoding [5]. The

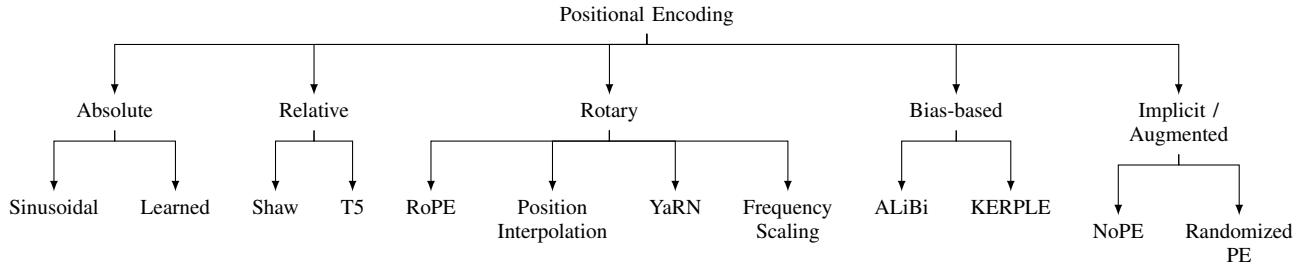


Fig. 1. A taxonomy of positional encoding methods for Transformer models.

method also integrates easily into existing transformer implementations. A limitation appears in tasks requiring retrieval across long distances. The linear penalty gradually suppresses attention assigned to distant tokens, producing a strong recency bias.

E. Methods for Extending RoPE

The widespread adoption of RoPE shifted recent research toward extending pretrained models instead of replacing the encoding entirely. Most approaches modify the positional representation during inference or apply brief fine-tuning to support longer contexts.

Position Interpolation rescales input position indices before applying RoPE [10]. Long evaluation sequences are compressed into the positional range observed during pretraining. The model therefore operates within familiar position values instead of extrapolating beyond them. Experimental results show stable performance after limited fine-tuning while extending the context window to several times the original length.

Frequency rescaling methods modify the RoPE frequency spectrum instead of compressing position indices. Liu et al. analyzed the relationship between rotary frequency and extrapolation stability through scaling laws [11]. Their work introduces the concept of a critical dimension, which determines where attention instability begins. Proper frequency scaling significantly increases the supported context length after additional training.

YaRN further improves RoPE extension by treating frequency components differently [12]. Low-frequency dimensions are interpolated, while high-frequency dimensions remain largely unchanged. The method also adjusts the attention softmax temperature to maintain stable optimization. Experimental results report long-context performance using substantially fewer training steps than Position Interpolation.

Related methods continue this direction. xPos introduces an exponential decay term within the rotary transformation while preserving relative position information [13]. The additional decay stabilizes long-distance attention and improves extrapolation over extended context windows.

F. Implicit, Randomized, and Kernelized Encodings

Recent research shifts away from standard deterministic and learned embeddings to explore alternative generalization

paradigms. Kazemnejad et al. investigated transformers operating without explicit positional encodings (NoPE) [14]. Their analysis reveals that decoder-only models trained with causal masking can implicitly learn positional relationships from the sequence structure, occasionally outperforming explicit absolute encodings on length generalization tasks.

To address the rigid boundaries of training windows, Ruoss et al. introduced Randomized Positional Encodings [15]. This scheme assigns randomized position indices to tokens during training, simulating the presence of much longer sequences. By training on sub-segmented, randomized index intervals, the self-attention mechanism learns to interpolate across a wider coordinate space, boosting the model’s capacity to handle extended inference contexts.

Another mathematically rigorous approach is Kernelized Relative Positional Embedding (KERPLE), proposed by Chi et al. [16]. KERPLE defines relative position embeddings through kernel functions, parameterizing the relative distance decays. This framework ensures that relative position representations remain bounded and positive-definite, enabling robust length extrapolation to arbitrary lengths during inference while maintaining stable attention statistics.

III. COMPARATIVE ANALYSIS

The positional encoding methods reviewed in Section II exhibit distinct trade-offs in representation strategy, parameter efficiency, and extrapolation capability. A systematic comparison of these paradigms reveals how structural design choices directly influence inference stability at extended context boundaries. Table I summarizes the major characteristics of the positional encoding methods reviewed in this paper.

A. Length Extrapolation Performance

Length extrapolation measures how well a transformer processes input sequences longer than those observed during training. Absolute encodings suffer from severe performance drops; learned embeddings restrict the model to a hard-coded coordinate limit, while sinusoidal encodings fail because out-of-distribution coordinate values disrupt learned attention dynamics [5]. Relative frameworks like Shaw’s relative representation and T5’s bucketed biases offer better translation invariance across moderate context expansions, though bucket boundaries eventually collapse under extreme limits [6], [7].

TABLE I
COMPARISON OF MAJOR POSITIONAL ENCODING METHODS

Method	Position Type	Extrapolation	Extra Parameters	Complexity	Representative Models
Sinusoidal	Absolute	Low	No	Low	Transformer
Learned	Absolute	Poor	Yes	Low	BERT
Shaw	Relative	Medium	Yes	Medium	Transformer
T5	Relative	Medium	Yes	Medium	T5
RoPE	Rotary	High	No	Low	LLaMA
ALiBi	Linear Bias	High	No	Very Low	Research
PI	RoPE Extension	Very High	No	Low	Long-context LLMs
YaRN	RoPE Extension	Very High	No	Low	LLaMA
Frequency Rescaling	RoPE Extension	Very High	No	Moderate	Long-context LLMs
KERPLE	Kernelized Bias	High	Yes	Moderate	Research
NoPE	Implicit	Medium-High	No	Very Low	Research (decoder-only)
Randomized PE	Absolute (Randomized)	High	No	Low	Research

RoPE achieves stronger extrapolation than absolute encodings because relative distance remains encoded as continuous vector rotations during query-key interactions [9]. However, high-frequency rotary dimensions exhibit phase-shifting instability over long ranges. Position Interpolation resolves this constraint by compressing coordinates into the pretraining coordinate boundaries, which requires brief fine-tuning to realign attention distributions [10]. YaRN and frequency rescaling optimize this compression by selectively scaling low- and high-frequency components, preventing local attention degradation and minimizing the computation needed for post-training adaptation [11], [12].

In contrast, ALiBi bypasses token embedding modifications entirely. Linear attention bias improves extrapolation by adding distance-based penalties directly to attention logits, ensuring stable softmax behavior across arbitrary sequence lengths [5]. This approach, however, induces a strict recency bias that can hinder performance in long-range multi-hop retrieval tasks where distant token associations are critical. Similarly, KERPLE leverages kernelized decays to optimize these distance-based penalties, balancing decay rate flexibility with mathematical stability [16].

A more surprising result comes from models that omit explicit positional encoding entirely. Because decoder-only architectures can implicitly recover ordering information from causal masking alone, Kazemnejad et al. found that NoPE outperforms ALiBi, RoPE, and learned absolute encodings on several downstream length-generalization tasks despite introducing no additional parameters or computation [14]. This finding suggests that explicit positional machinery, while useful, is not strictly necessary for length generalization in causal transformers, and that attention stability rather than the specific encoding mechanism may be the more fundamental driver of extrapolation performance.

B. Computational Cost

Evaluating computational overhead requires analyzing training parameters, execution latency, and memory footprints. Sinusoidal encodings demand zero parameter overhead and rely on static trigonometric computations during sequence initialization. Learned absolute embeddings introduce param-

eters proportional to the context limit but maintain negligible runtime overhead via direct table lookups.

Relative techniques increase calculation complexity. Shaw’s relative representations require calculating pairwise offset vectors, which scales poorly in memory usage. T5 relative bias minimizes this constraint by mapping offsets to scalar buckets, though index bucketing still adds operational overhead before self-attention calculation.

RoPE rotates query and key vectors using element-wise multiplication, introducing negligible processing latency while keeping additional parameter counts at zero [9]. Linear attention bias improves extrapolation while maintaining extremely low implementation complexity, as static matrix additions are applied directly to raw attention scores. Position Interpolation, YaRN, and frequency rescaling preserve the computational profile of their base rotary models, requiring no changes to the self-attention kernel architecture.

C. Practical Considerations

Practical selection depends on model lifecycle constraints, particularly the cost of pretraining. RoPE currently serves as the default option for modern open-source foundations like LLaMA because of its balance of zero parameter overhead and high contextual representation capacity [2]. Consequently, post-training methods like YaRN and Position Interpolation are widely adopted to extend existing rotary checkpoints without restarting pretraining.

ALiBi presents a highly effective alternative when training new architectures from scratch. Linear attention bias simplifies implementation and provides robust extrapolation out-of-the-box, though integrating it into existing rotary-based models is impractical due to weights incompatibility. For systems requiring strict mathematical guarantees on position decay, kernelized options like KERPLE provide continuous relative representations, whereas randomized positional encodings offer an inexpensive training-time modification to make standard architectures length-agnostic.

IV. CONCLUSION

This paper presented a comparative study of positional encoding methods designed for length extrapolation in transformer architectures. We structured the major approaches

into absolute, relative, rotary, and bias-based paradigms. The review traced the evolution from early static sinusoidal vectors to modern continuous rotations and attention-level distance penalties, illustrating how each architectural design represents sequence ordering.

Our comparison highlights that absolute embeddings fail to generalize beyond pretraining limits. Relative and bias-based techniques, such as T5 and ALiBi, achieve robust length extrapolation but introduce operational overhead or strict recency biases that can degrade long-range dependency retrieval. Rotary positional representations and their recent post-training scaling variants currently offer the most effective path for extending context windows in massive language models, balancing computational efficiency with performance preservation.

Future research directions should focus on addressing the trade-offs of existing scaling techniques, particularly by investigating hybrid encodings that combine relative coordinate properties with implicit positional indicators. Additionally, further studies are needed to evaluate length generalization in non-causal attention models and to establish unified, downstream task-based benchmarks that look beyond simple perplexity evaluations.

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